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EXPLAINED

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Why Has the U.S. Deployed 'Space Weapons'?

Why has U.S. deployed 'space weapons'?

What is a space control 'weapon'? What has the U.S. said? Are there treaties limiting weapons deployments in space?

Vandevan Mukumb

The story so far:
On September 14, U.S. Air Force secretary Troy Klum said the country has "space control" weapons deployed in earth orbit to defend U.S. forces against "hostile adversary action". The disclosure drew backlash from Russia and China, among other quarters.

What is a 'space control' weapon?
As per the U.S. Space Force, 'space control' refers to military activities required to control the space domain. In 2025, the then Space Force chief Gen. Chance Saltzman had said such a device could possess both 'kinetic and non-kinetic' capabilities and could disrupt or degrade an adversary's capabilities.
'Kinetic' generally refers to physically damaging an object through impact, such as by smashing into it. Non-kinetic methods include electromagnetic interference that can jam or spoof a satellite's communications, and directed energy systems that interfere with its sensors.
In fact, the current Space Force war-fighting framework recognises three kinds of counter-space operations: orbital, electromagnetic, and cyberspace.

What has the U.S. deployed?
We do not know. We also do not know how many instances of the system exist, what they are capable of, which orbits they are in, and whether they are kinetic or non-kinetic. Mr. Klum had said that while disclosing the weapon's (or weapons') existence would contribute to deterrence, revealing their operational details would subtract from the deterrence. There is also no indication whether these weapons are the same as the space-based interceptors the U.S. government has planned for the 'Golden Dome' missile defence programme. These interceptors are still undergoing tests.

What will the weapons defend against?
Mr. Metink has referred to 'hostile adversary action'. Incumbent Space Force chief Gen. Douglas Schies has also referred to 'space-enabled attacks'.
In other instances over the years, U.S. officials have reportedly singled out Russian and Chinese counter-space capabilities as reasons for their government to develop similar capabilities of its own, to protect its own satellites. Russia and China have also demonstrated sophisticated satellite manoeuvres in orbit.
For example, in 2021, two Russian satellites manoeuvred close to a U.S. government satellite, prompting the Space Command to call the behaviour potentially threatening. In 2022, the Chinese Shijian-2 satellite used a robotic arm to tow a defunct Chinese geostationary satellite into a graveyard orbit.
However, we do not know the circumstances in which the U.S. will use its weapons as the government has not published the rules of engagement for orbital weapons.

Is it legal to have weapons in orbit?
The relevant instrument is the Outer Space Treaty, which entered into force in 1967 during the Cold War and the nuclear arms race between the U.S. and the Soviet Union. It prohibits in party states, which include the U.S., Russia, and China today, from placing nuclear weapons or other weapons of mass destruction in orbit or stationing them elsewhere in outer space. It also obligates states from setting up military bases and conducting weapons tests on the moon and other celestial bodies.
However, the treaty does not prohibit states from placing conventional weapons in orbit. The treaty also says all activities in outer space must comply with international law, which means other bodies of international law governing the use of force and armed conflicts can be applied to military operations in orbit. As the U.S. government has not disclosed what its orbital weapons are capable of, it is not possible to say whether it violates specific treaty conditions.

Can other countries prevent unilateral action?
There is no international body to order the U.S. to remove its weapons from orbit. However, there are various diplomatic and legal avenues through which other countries can seek more information and negotiate restraints. For example, Article IX of the treaty requires states to have "due regard" for the interests of other parties, which a (party) state can invoke if it believes U.S. activities in orbit could harm its own peaceful use of space. States can also endeavour to draft new orbital governance rules through the United Nations process on the Prevention of an Arms Race in Outer Space, whose third substantive session is scheduled for November. Countries can also raise the issue before the UN Security Council or the General Assembly as a threat to international peace and security, under Article 25 of the UN Charter.

Basics

Outer space is treated in international law as a global commons, a domain open to all states but owned by none.

Four Phases

Cold War Era Treaty → Institutional Consolidation → Commercialisation → Weaponization

Cold War Era Treaty (1957 - 1979)

The launch of Sputnik 1 by the Soviet Union on 4 October 1957 marked the beginning of the space age and ignited a Cold War competition between the United States and the Soviet Union

The United Nations General Assembly established the Committee on the Peaceful Uses of Outer Space (COPUOS) as a permanent body in 1959. Headquartered in Vienna and supported by the UN Office for Outer Space Affairs (UNOOSA)

Basics

Outer Space Treaty, 1967: Often called the Magna Carta of space law, it was opened for signature on 27 January 1967 and entered into force on 10 October 1967.

Its key principles are

- **Freedom of exploration and use for the benefit of all countries (Article I);**
- **Non-appropriation of outer space, the Moon, and other celestial bodies by claim of sovereignty (Article II);**
- **Conduct of space activities in accordance with International law (Article III);**
- **A ban on placing weapons of mass destruction in orbit and a requirement that the Moon and celestial bodies be used exclusively for peaceful purposes (Article IV);**
- **State responsibility for national space activities, including those carried out by private or non-governmental entities (Article VI);**
- **State liability for damage caused by space objects (Article VII).**
- **Astronauts are described as envoys of mankind.**

Basics

Rescue Agreement, 1968: Obligates states to render assistance to astronauts in distress and to safely and promptly return them, along with any space objects that land outside the territory of the launching state.

Liability Convention, 1972: Establishes absolute liability for a launching state for damage caused by its space object on the surface of the Earth or to aircraft in flight, and fault-based liability for damage caused elsewhere, such as in outer space.

Registration Convention, 1976: Requires launching states to maintain a national registry and to furnish details of space objects to a central register maintained by the UN Secretary-General, aiding jurisdiction, tracking, and accountability.

Moon Agreement, 1979: Introduced the principle that the Moon and its natural resources are the common heritage of mankind and called for an international regime to govern future exploitation of lunar resources. The Agreement entered into force in 1984 but has never been ratified by any major spacefaring power; the United States, Russia, and China are not parties, and India signed in 1982 but did not ratify. It therefore remains largely ineffective as a governing instrument, leaving the question of space resource rights unresolved for decades.

India is a signatory to all five of these treaties but has ratified only four. India did not ratify Moon agreement.

Basics

Phase II: Institutional Consolidation and Soft Law (1980s to 1990s)

As consensus among spacefaring and non-spacefaring states became harder to achieve, treaty making gave way to non-binding declarations and principles adopted by the UN General Assembly.

Phase III: Commercialisation and the Governance Gap (2000s to 2010s)

The rise of private spaceflight, exemplified by the founding of SpaceX in 2002, drove rapid growth of a commercial launch industry that existing treaties had not anticipated.

With the Moon Agreement unratified by major powers, individual states began legislating unilaterally on space resource rights: the United States enacted the Commercial Space Launch Competitiveness Act, 2015, permitting US citizens to own resources they extract from celestial bodies

Phase IV: Contemporary Governance Challenges (2020s Onward)

Weaponization and Anti-Satellite Tests

The Outer Space Treaty bans only weapons of mass destruction in orbit; conventional anti-satellite (ASAT) weapons remain unregulated by any binding instrument.

On 27 March 2019, India conducted Mission Shakti, its first ASAT test, destroying its own defunct Microsat-R satellite at an altitude of approximately 283 km using the DRDO-developed Prithvi Delivery Vehicle Mark II.

India became the fourth country, after the United States, Russia, and China, to demonstrate this capability; the low test altitude was chosen to minimise long-lived debris.

Basics

The underlying hazard is the Kessler Syndrome, a scenario proposed by NASA scientist Donald Kessler in 1978, in which debris density in low Earth orbit becomes high enough to trigger a self-sustaining cascade of collisions that could render affected orbits unusable for generations.

The Artemis Accords, 2020

Launched in October 2020 and led by NASA together with the US Department of State and founding partner nations, the Artemis Accords are a non-binding set of principles for civil exploration of the Moon, Mars, comets, and asteroids, explicitly grounded in the Outer Space Treaty

Explicit allowance for the extraction and utilisation of space resources.

India signed the Artemis Accords in June 2023, a departure from its traditional preference for formal, multilateral, treaty-based governance

Current Context :On September 14, U.S. Air Force secretary Troy Meink said the country has 'space control' weapons deployed in earth orbit to defend U.S. forces against 'hostile adversary action' : the disclosure drew backlash from Russia and China.

Basics

"Mission Shakti" (DRDO) is the name given by India to

This question was previously asked in
**UPSC CAPF AC 2024 Official
Paper-I (Held on 4 August 2024)**

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2. Air-to-Air Missile
3. Anti-Satellite Missile
4. Nuclear-Powered Attack Sub-marine

Experts Warn of Devastating El Niño Impacts on India

Experts warn of devastating El Niño impacts on India

The Hindu Bureau
BENGALURU

As many as 188 of India's scientists and ecologists jointly published an appeal to fellow researchers and field practitioners to prepare for the effects of El Niño – a global weather phenomenon that dampens the monsoon and leads to extreme hot and dry conditions in the subcontinent.

In the Pacific Ocean where El Niño originates, temperatures have steadily risen above the 0.5 degree Celsius threshold, reaching 2.6 degree Celsius in August.

Projections report possible anomalies of up to 4 degrees Celsius, which is higher than at any time in the past 1,000 years, based on reconstructions from coral cores, they said.

El Niño started in May

and its main effects are expected to be felt in the northern hemisphere (including India) from November onwards into the next year, they said, suggesting a 'Super' El Niño. Over the coming months, the signatories warned, India's natural systems are likely to experience a range of potentially devastating impacts as a result of El Niño.

"These include widespread bleaching and mortality of coral, an increase in wildfires, and the death of large rainforest trees. With a high probability of drought and extreme heat conditions, this year's El Niño is likely to be catastrophic for India's farmers, fishers and shepherds as well. In addition though, more subtle impacts on India's nature – including delayed fruit production or



Projections report temperatures rising by up to 4 degrees Celsius based on reconstructions from coral cores. SPECIAL ARRANGEMENT

changes in animal behaviour – could be equally important, but may be much less visible unless you are specifically looking for them," they said.

The "Call to Action" urges scientists to take advantage of the warning signs and mobilise as a community to document and study the impact of this event on India's nature, arguing that

better science and documentation are essential to preparing India for a climate resilient future.

Jayashree Ratnam, scientist, National Centre for Biological Science, said in a release: "The coming year will be a litmus test for tropical forests and trees in India. Huge forest trees may appear indestructible, but in the Western Ghats,

many of them are already near the limit of the heat they can tolerate. Large trees are especially susceptible to extended droughts. Trees in cities are also vulnerable because of the urban 'heat island' effect. It is vital to understand how the coming El Niño will impact trees and other plants – as these are the foundations on which all other life depends."

'Scary projections'

"I have witnessed the devastating impacts of El Niños on India's coral reefs before, and it is heart-breaking. Coral reefs can be remarkably resilient but they need at least a decade after every major die-back of coral to show significant recovery. The last major El Niño was in 2024, and barely two years after that event, the projected scale

and intensity of the heat wave that is currently headed our way is truly scary," said Rohan Arthur, marine biologist, Nature Conservation Foundation.

Roxy Mathew Koll, climate scientist, Indian Institute of Tropical Meteorology, also said India's monsoon season is closing with a rainfall deficit, but the impacts may not end there. "They can creep into the seasons ahead and cascade through depleted soil moisture and water storage, land and marine heatwaves, stressed crops and livelihoods, forest fires, coral bleaching and less visible ecological changes. We must document these changes as they unfold, but also safeguard the ecosystems that remain and reduce the pressures already weakening them," he said.

Concept

Experts Warn of Devastating El Niño Impacts on India

El-Nino Impact on India

Monsoon Rainfall : El Nino is historically associated with a weaker south west monsoon over India,

Agriculture and Food Security : A weak or delayed monsoon directly affects kharif crop sowing and yields, particularly rice, pulses, oilseeds, sugarcane, and cotton, most of which remain heavily dependent on monsoon rainfall despite expanding irrigation

Water Resources and Hydropower : Delayed or deficient rainfall lowers reservoir inflows, affecting both irrigation supply and hydropower generation

Fisheries : Along the Indian coast, particularly Kerala, warming of coastal waters during El Nino years has been associated with declines in the oil sardine catch, since sardine populations are sensitive to sea surface temperature and upwelling intensity

Monetary Policy and Inflation Management : Because food articles carry a significant weight in India's Consumer Price Index

What Is the India-Pakistan Conflict at Sea About?

What is the India-Pakistan conflict at sea about?

What happened on September 15 in the North Arabian Sea? What kind of vessels were involved? Was there any damage to the vessels? How did India react to the incident? What is Pakistan's version? What is Article 10 of the 1991 Agreement that India alleged Pakistan violated?

Saurabh Trivedi

The story so far:

A Pakistan Navy vessel, PNS Hunain, collided with an Indian Navy warship, INS Kolkata, in the North Arabian Sea on September 15 after, according to India, the Pakistani ship made a dangerous overtaking manoeuvre at high speed. India summoned Pakistan's Charge d'Affaires and cited violations of Article 10 of the 1991 India-Pakistan agreement, which requires naval vessels to maintain separation of three nautical miles. Pakistan rejected India's allegations. INS Kolkata suffered no major damage and remains at sea. PNS Hunain was damaged and subsequently returned to harbour.

What are INS Kolkata and PNS Hunain?

The two vessels involved are different classes of warships. INS Kolkata is the lead ship of the Indian Navy's Project 15A Kolkata-class guided missile destroyers. Built by Mazagon Dock Shipbuilders in Mumbai, the 164-metre-long

New Delhi has advised Pakistan to ensure that its military units exercise due care and comply with relevant bilateral agreements to prevent a recurrence.

warship has a full-load displacement of about 7,400 tonnes and is capable of sailing at 30-knot speed. It was commissioned into the Indian Navy in August 2014. INS Kolkata is designed primarily as a high-end surface combatant, with long-range air-defence, anti-surface, and anti-submarine capabilities. The class is equipped to operate helicopters and carries the BrahMos supersonic cruise missile as its principal anti-surface strike weapon.

PNS Hunain (F-273) is an offshore patrol vessel. It was built by Damen Shipyards in Romania and commissioned in July 2024. It is the first of Pakistan's larger OPV 2600-based follow-on vessels to the Yarmook-class patrol ships. Damen says the OPV 2600 design is intended for maritime security and military operations, including sustained operations in contested environments. It is approximately 98 metres long, has a displacement of about 2,600 tonnes, and is capable of speeds of up to about 24 knots. Its role includes maritime surveillance, patrol and security missions. Pakistan describes the vessel as a multi-purpose platform equipped with electronic-warfare, anti-ship and anti-air warfare capabilities, as well as self-protection and terminal defence systems. It can also embark a multirole helicopter.

What does India say happened?

According to the Indian account, INS Kolkata was conducting routine surveillance when PNS Hunain approached at high speed and manoeuvred dangerously close to the Indian ship. On September 18, a spokesperson of the Ministry of External Affairs said India had examined Pakistan's response to the diplomatic protest and rejected its allegations. He said the Pakistani ship had conducted a "dangerous overtaking manoeuvre" that resulted in the collision.

India has maintained that the manoeuvre violated both the 1991 bilateral agreement and the International Regulations for Preventing Collisions at Sea. The Ministry said INS Kolkata suffered no major damage and remained

deployed at sea.

Pakistan has given a different account, alleging that the Indian warship carried out a provocative manoeuvre during the encounter. The precise sequence of movements immediately before the collision therefore remains disputed.

What is Article 10 of the 1991 India-Pakistan agreement?

Article 10 of the Agreement between India and Pakistan on Advance Notice on Military Exercises, Manoeuvres and Troop Movements, signed on April 6, 1991, lays down a specific maritime safety provision. It states that naval ships and submarines of the two countries should not approach within three nautical miles of each other while operating in international waters, "so as to avoid any accident." Three nautical miles is about 5.6 km.

The provision is part of a broader military confidence-building framework intended to reduce the risk of miscalculation between the two countries. It is a bilateral requirement, not a universal international maritime rule applicable to warships of all countries.

Has this happened before?

The September 2026 incident is the second reported India-Pakistan naval collision involving the 1991 framework. In June 2011, Pakistani warship PNS Babur came close to Indian Navy frigate INS Godavari during anti-piracy operations in the Gulf of Aden, causing minor damage. India lodged a protest citing navigational-safety violations.

What happens next?

India has summoned Pakistan's Charge d'Affaires in New Delhi and lodged a strong protest. It has also instructed its Charge d'Affaires in Islamabad to raise the matter with Pakistan's Foreign Ministry.

New Delhi has advised Pakistan to ensure that its military units exercise due care and comply with relevant bilateral agreements to prevent a recurrence. The fact that the collision occurred in international waters does not place the two navies outside agreed rules of conduct.

For now, the key unresolved question is the exact sequence of manoeuvres that brought the two vessels into close proximity.



Ministry of External Affairs official spokesperson Randhir Jaiswal during a briefing. PTI

Concept

What Is the India-Pakistan Conflict at Sea About?

Advance Notice on Military Exercises, Manoeuvres and Troop Movements

The Agreement between India and Pakistan was signed in April 1991.

Major military exercises close to each other's territory should be avoided. If held, they must follow set regulations and the other side must be informed.

Advance notice periods: 15 days for certain air and naval exercises, 60 days for corps-level exercises, 90 days for army-level exercises. Either side can seek clarification on assembly of forces, direction, extent and duration of an exercise.

Naval definition: A major naval exercise involves six or more ships of destroyer or frigate size and above, exercising together and crossing into the other's EEZ.

Article 10: Naval ships and submarines of the two countries must not close within three nautical miles of each other in international waters, to avoid accidents. This is the provision India says was violated.

Aircraft rules: Combat aircraft must not fly within 10 km of each other's airspace, including Air Defence Identification Zones.

Concept

What Is the India-Pakistan Conflict at Sea About?

Grey-zone warfare refers to a form of conflict characterised by actions that fall below the threshold of conventional war but are intended to achieve strategic objectives through ambiguity, deniability, and coercion.

In grey-zone warfare, adversaries employ tactics such as cyberattacks, economic coercion, and proxy conflicts to achieve their goals without directly engaging in open warfare.

It blurs the lines between peace and conflict, posing significant challenges to international security and stability.

Explain the structure of the Parliamentary Committee system. How far have the financial committees helped in the institutionalization of Indian Parliament?

Thank You,